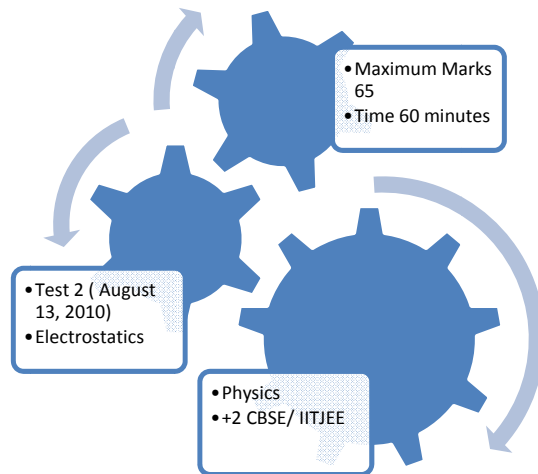


Name
Class
R. No.



Instructions

1. The test contains five questions

Q1. A positive point charge $50\mu C$ is located in the plane xy at the point with radius vector $\vec{r}_0 = 2\hat{i} + 3\hat{j}$, where $\hat{i}$ and $\hat{j}$ are the unit vectors of the x and y axes. Find the vector of the electric field strength $\vec{E}$ and its magnitude at the point with radius vector $\vec{r} = 8\hat{i} - 5\hat{j}$. Here r_0 and r are expressed in metres . 10 Marks

Q2 . A thin half-ring of radius $R = 20$ cm is uniformly charged with a total charge $q = 0.70$ nC . Find the magnitude of the electric field strength at the curvature centre of this half-ring.

10 Marks

Q3. A system consists of a thin charged wire ring of radius R and a very long uniformly charged thread oriented along the axis of the ring, with one of its ends coinciding with the centre of the ring. The total charge of the ring is equal to q . The charge of the thread (per unit length) is equal to A . Find the interaction force between the ring and the thread.

10 Marks

Q4. A thin nonconducting ring of radius R has a linear charge density $\lambda = \lambda_0 \cos \phi$, where λ_0 is a constant, ϕ is the azimuthal angle. Find the magnitude of the electric field strength
(a) at the centre of the ring;
(b) on the axis of the ring as a function of the distance x from its centre. Investigate the obtained function at $x \gg R$

25 Marks

Q5. A very long straight uniformly charged thread carries a charge λ per unit length. Find the magnitude and direction of the electric field strength at a point which is at a distance y from the thread and lies on the perpendicular passing through one of the thread's ends.

10 Marks